K. J. Somaiya School of Engineering

Batch: A3 Roll No.: 16010324044
Experiment ~~/ assignment / tutorial No~~. 05
Grade: AA / AB / BB / BC / CC / CD /DD

Signature of the Staff In-charge with date

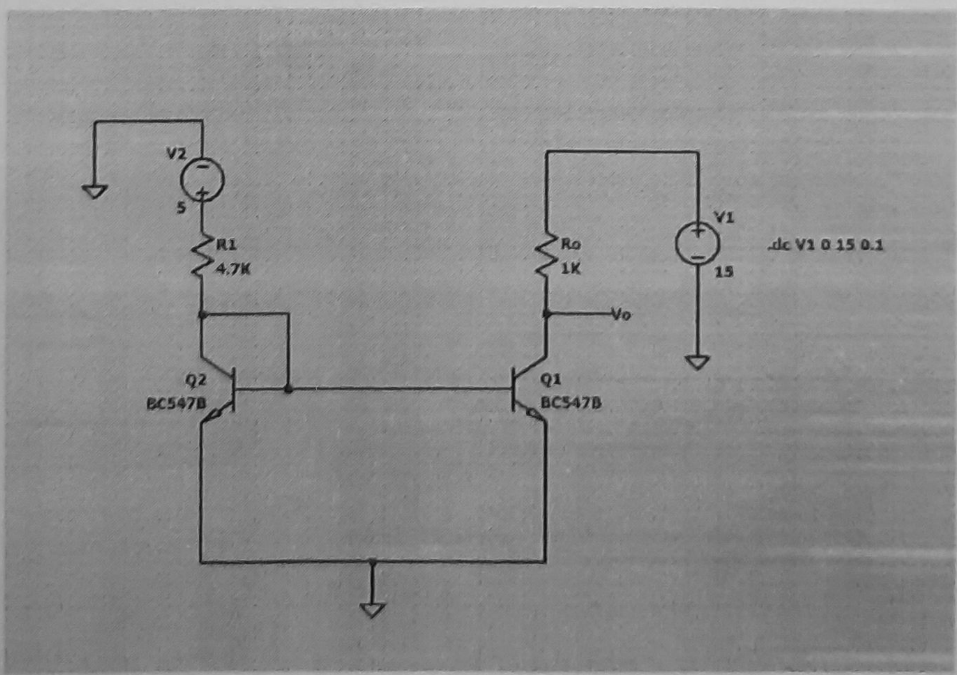
TITLE: Performance evaluation of different types of Constant Current Source
AIM: To evaluate the output resistance of different types of constant current source using LtSpice.
OUTCOME: Students will be able to understand the basics of analog Integrated Circuits

PROCEDURE:

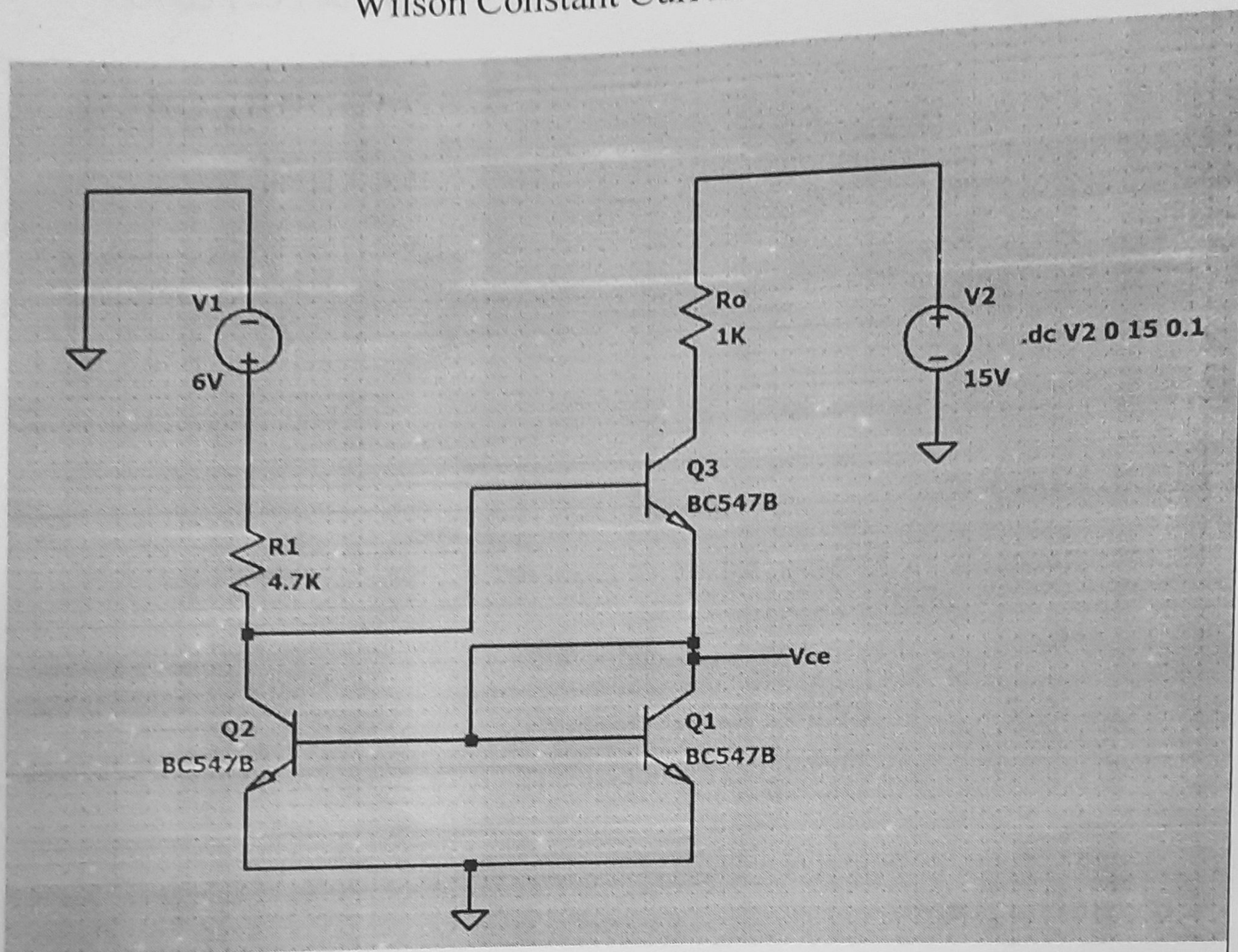
1. Implement the basic two transistor constant current biasing circuit using LtSpice.
2. Adjust Vcc such that $I_{ref} = 1$ mA.
3. Find the values of Io and V_{CE} (collector voltage) as Vs is varied from 15V to 0V as per the observation table I.
4. Find the dynamic output resistance by plotting Io versus V_{CE} curve.
5. Repeat above procedure in workspace of Ltspice for Wilson and Widlar constant current biasing circuit.
6. Comment on the output resistance offered by each circuit.

Circuit Diagram

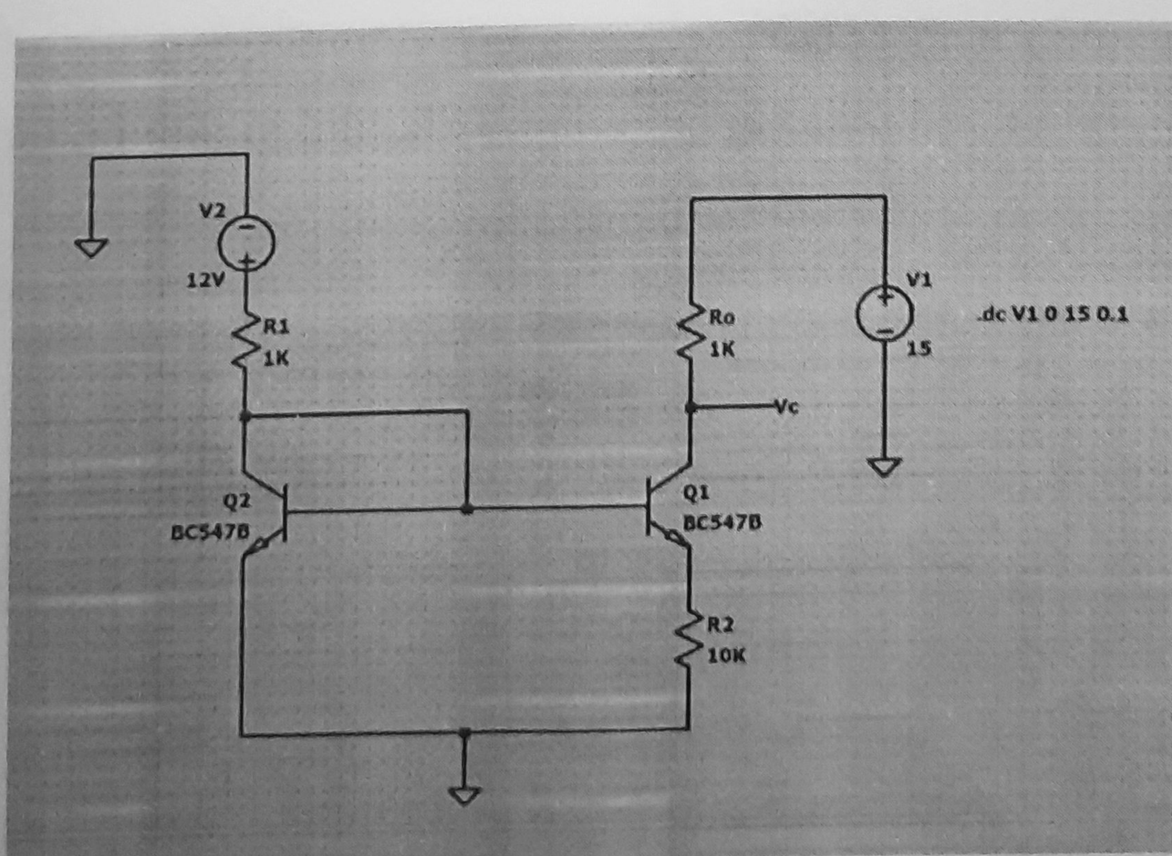
Basic Two Transistor Constant Current Source



Wilson Constant Current Source

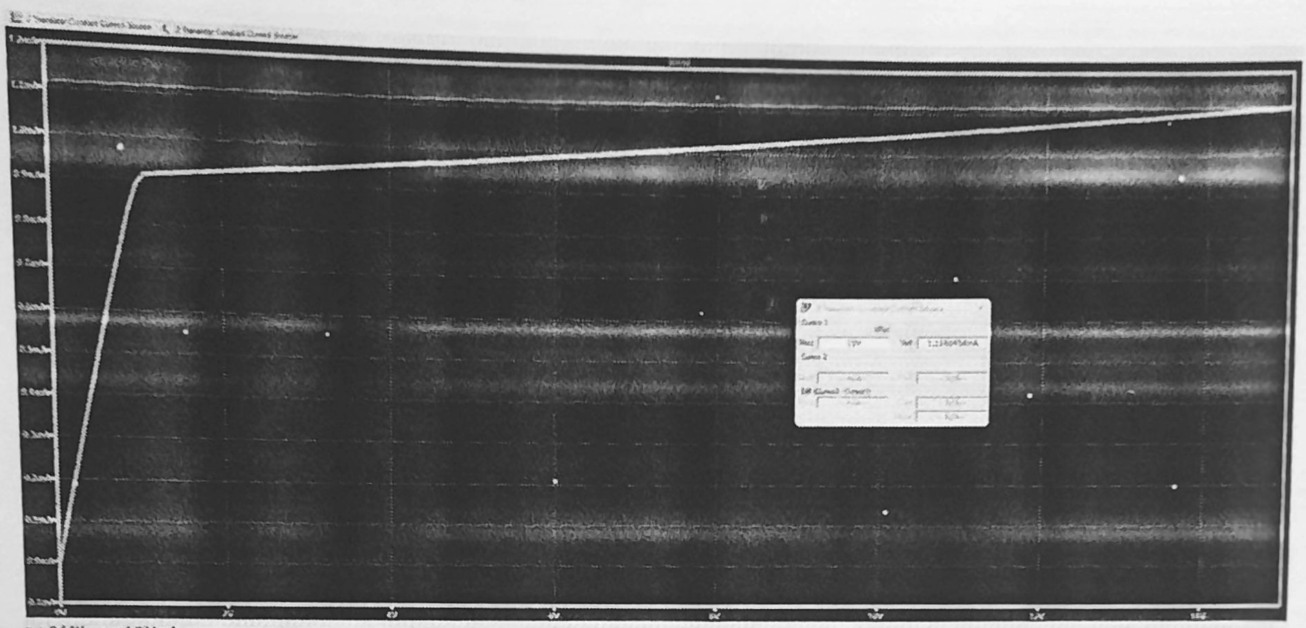


Widlar Constant Current Source

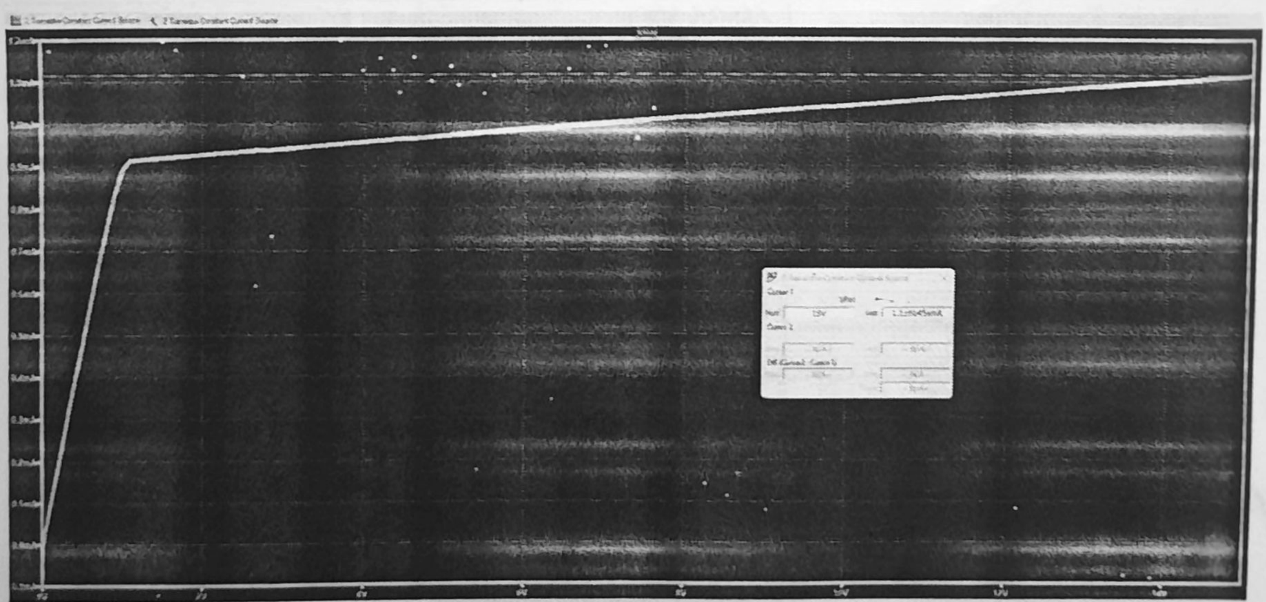


Graphs

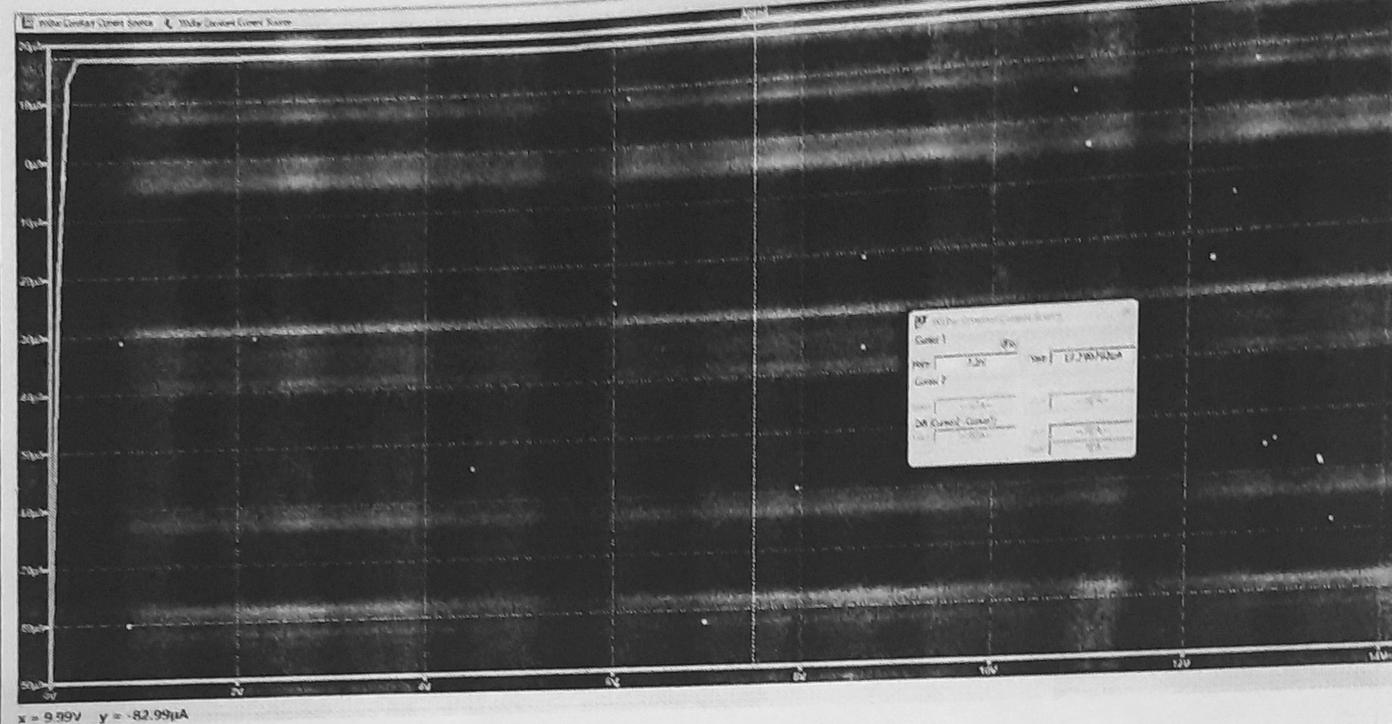
Basic Two Transistor Constant Current Source



Wilson Constant Current Source



Widlar Constant Current Source



Observation Table I

	Simulation	
	Output Resistance (ro)	Output Current (Io)
Basic Two transistor constant current source	69.35 KΩ	1.116 mA
Wilson constant current source	6.055 MΩ	996.95 uA
Widlar constant current source	31.¥56 MΩ	18.03 uA

Mathematical Analysis and Theoretical Calculations

$$\text{Slope } m = \frac{I_{02} - I_{01}}{V_{CC2} - V_{CC1}} \Big\} \text{ LT Spice}$$

$$\therefore r_0 = \frac{1}{m}$$

① Basic 2-Transistor constant current source:

$$r_0 = 1/(1.44)(10^{-5}) = 69.35 \, k\Omega$$

② Wilson Constant Current Source: $r_0 = \frac{1}{(1.65)10^{-7}} = 6.055$

③ Wildar Current Constant Source: $r_0 = \frac{1}{(3.48)10^{-8}} = 31.7$

Conclusion: Types of Constant Current Sources explained various current mirroring techniques used to replicate current across while maintaining stability against load ~~vario~~ variations. Using LT Spice, three specific circuits were simulated to analyze their ~~d~~ output characteristics. The result demonstra that ~~wid~~ widlar & wilson current source provide a higher output resistance (r_o) that the basic two-transistor model, effectively counteracting Early Effect & improving performance.

Signature of faculty